

files at least would need to be updated.

- `/etc/ldap.conf`: For information about the central LDAP server and its settings
- `/etc/krb5.conf`: For information about the central Kerberos infrastructure
- `/etc/sss/sss.conf`: To configure the system security services daemon (sssd), which is the daemon responsible for retrieving and caching user information and authentication information
- `/etc/nsswitch.conf`: To indicate to the system which user information and authentication services should be used
- `/etc/pam.d/*`: To configure how authentication should be handled for various services
- `/etc/openldap/cacerts`: To store the root certificate authorities (CA) that can validate the SSL certificates used to identify LDAP servers

The sssd daemon will need to be enabled and started before the system can use it. With so many files and services to configure, a mistake can easily be made. Red Hat Enterprise Linux 7 (RHEL 7) ships with a suite of tools to automate these configurations, called Authconfig, which consists of three related tools that can all perform the same actions.

- **Authconfig**: This is a command line tool that can be used to automate configurations across a number of systems. The commands used with Authconfig tend to be very long, with multiple options being passed in. This tool can be installed using the Authconfig package.
- **Authconfig-tui**: The interactive version of Authconfig uses a menu-driven text interface which can be used over SSH. This tool can be installed using the Authconfig package.
- **Authconfig-gtk**: This version launches a graphical interface. It can be launched as system-config-authentication. This tool is installed using the Authconfig-gtk package.

The necessary LDAP parameters

To connect to a central LDAP server for user information, Authconfig needs a number of settings:

- The host name of the LDAP server.
- The base DN (Distinguished Name) of the part of the LDAP tree where the system should look for users. This typically looks something like `dc=example, dc=com`. This information is known to LDAP server administrators.
- If SSL/TLS is used to encrypt communications with the LDAP server, a root CA certificate that can validate the certificate is offered by the LDAP server.

Note: A system will also need some extra packages installed to provide LDAP client functionality. Installing sssd will provide all the necessary dependencies.

The required Kerberos parameters

To configure a system to use a centralised Kerberos

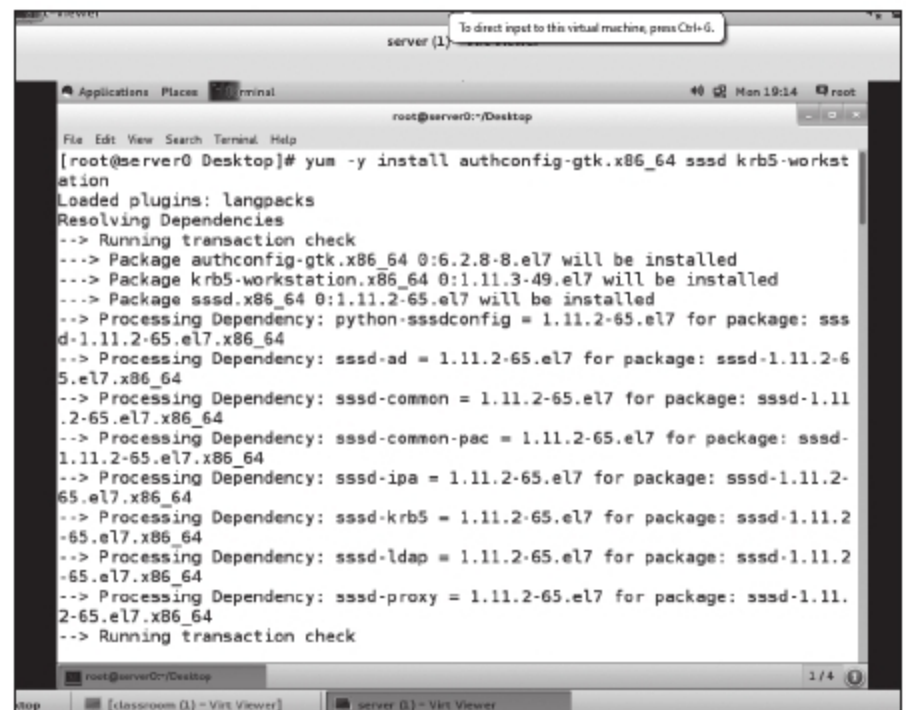


Figure 1: LDAP Kerberos packages are installed

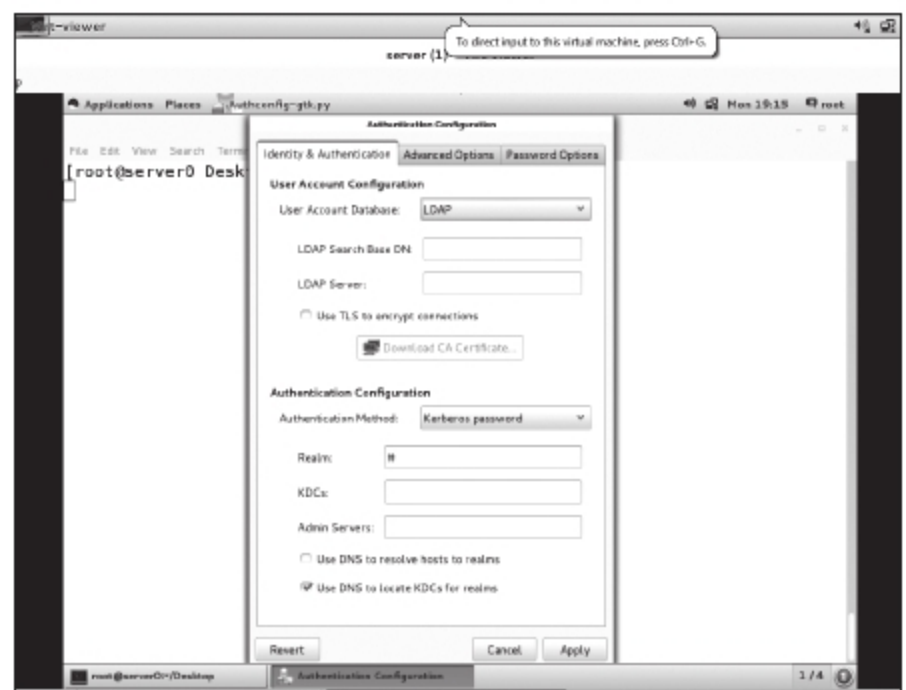


Figure 2: Main Authconfig-gtk window

system for user authentication, Authconfig will need the following settings:

- The name of the Kerberos realm to use. A Kerberos realm is a domain of machines, all of which use a common set of Kerberos servers and users for authentication.
- One or more key distribution centres (KDC). This is the host name of your Kerberos server.
- The host name of one or more admin servers. This is the machine that clients will talk to when they want to change their password, or perform other user modifications. This is typically the same as the primary KDC, but it can be a different machine.

In addition, a systems administrator can specify if the DNS should be used to look up the realm to use for a specific host name and to automatically find the KDCs and admin servers. An extra package can be installed to help debug Kerberos issues and to work with Kerberos tickets from the command line:

`krb5-workstation`